

Lecture 11: Multivariate Volatility Modeling

Outline

11.1 Basic Concepts

11.2 Two Simple Models (EWMA & BEKK)

11.3 Dynamic Conditional Correlation (DCC) Models

11.4 Copula-Based Models

11.5 Examples

11.6 VaR and ES via Multivariate Modeling

11.1 Basic Concepts

How do the correlations between asset returns change/evolve over time?

- Focus on two asset return series for ease of demonstration:

$$\mathbf{r}_t = \begin{bmatrix} r_{1,t} \\ r_{2,t} \end{bmatrix}$$

- In general: $\mathbf{r}_t$ is of dimension k .
- Data: $\mathbf{r}_1, \dots, \mathbf{r}_T$.
- F_{t-1} : the information set up to time $t - 1$.
- Partition $\mathbf{r}_t$ as:

$$\mathbf{r}_t = \boldsymbol{\mu}_t + \mathbf{a}_t, \quad \mathbf{a}_t = \Sigma_t^{1/2} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim (\mathbf{0}, I_k),$$

where

$$\boldsymbol{\mu}_t = \mathbb{E}(\mathbf{r}_t | F_{t-1}) = \begin{bmatrix} \mu_{1,t} \\ \mu_{2,t} \end{bmatrix} = \text{conditional mean}$$

$$\Sigma_t = \text{Cov}(\mathbf{a}_t | F_{t-1}) = \begin{bmatrix} \sigma_{11,t} & \sigma_{12,t} \\ \sigma_{21,t} & \sigma_{22,t} \end{bmatrix} = \text{conditional covariance}$$

- Concisely: $\mathbf{r}_t | F_{t-1} \sim (\boldsymbol{\mu}_t, \Sigma_t)$.
- Restrictions: none on $\boldsymbol{\mu}_t$, but Σ_t must be **symmetric** and **positive definite** (for all t):

$$\sigma_{12,t} = \sigma_{21,t}, \quad \sigma_{11,t} > 0, \quad \sigma_{22,t} > 0, \quad \sigma_{11,t}\sigma_{22,t} - \sigma_{12,t}^2 > 0.$$

- Dynamic (conditional) correlation between $\mathbf{r}_{1,t}$ and $\mathbf{r}_{2,t}$:

$$\rho_{12,t} = \frac{\sigma_{12,t}}{\sqrt{\sigma_{11,t}\sigma_{22,t}}}.$$

- Complications about Σ_t :

(i) positive definite requirement is difficult to enforce;

(ii) computational burden increases rapidly with k since there are $k(k+1)/2$ independent components (curse of dimensionality).

- Additional decompositions:

- Let $V_t = \text{Diag}(\sqrt{\sigma_{11,t}}, \dots, \sqrt{\sigma_{kk,t}})$ be the diagonal matrix consisting of just the diagonal elements of Σ_t . E.g., when $k = 2$ we have:

$$V_t = \begin{bmatrix} \sqrt{\sigma_{11,t}} & 0 \\ 0 & \sqrt{\sigma_{22,t}} \end{bmatrix}.$$

- Let $R_t = [\rho_{ij,t}]$ be the conditional correlation matrix, so that:

$$\Sigma_t = V_t R_t V_t, \quad \text{and} \quad R_t = V_t^{-1} \Sigma_t V_t^{-1}.$$

- Coverage: Ch. 10 of Tsay's AFTS (2010), and Ch. 7 of Tsay's MTS (2014). These both rely on the R package **MTS**.

11.2 Simple Models

Starting from the WN residuals:

$$\hat{\mathbf{a}}_t = \begin{bmatrix} \hat{a}_{1,t} \\ \vdots \\ \hat{a}_{k,t} \end{bmatrix} = \mathbf{r}_t - \hat{\boldsymbol{\mu}}_t, \quad (1)$$

where the conditional mean can be estimated via, e.g., a VAR model, we can fit the following (simple) models to capture the volatility Σ_t .

EWMA. A crude estimate of Σ_t is to use the (unconditional) empirical covariance estimate of the innovations:

$$\hat{\Sigma}_t = \frac{1}{t-1} \sum_{i=1}^{t-1} \mathbf{a}_i \mathbf{a}_i', \quad (2)$$

which assigns equal weight of $1/(t-1)$ to all observations. The **exponentially weighted moving average (EWMA)** model, improves on this by using a weighted combination of the latest observations and the previous estimate:

$$\hat{\Sigma}_t = (1 - \lambda) \mathbf{a}_t \mathbf{a}_t' + \lambda \hat{\Sigma}_{t-1},$$

where $0 < \lambda < 1$. Iterating this leads to the infinite sum representation:

$$\hat{\Sigma}_t = (1 - \lambda) \sum_{i=1}^{\infty} \lambda^{i-1} \mathbf{a}_{t-i} \mathbf{a}_{t-i}'.$$

In order to estimate λ , we assume $\mathbf{a}_t = \mathbf{r}_t - \boldsymbol{\mu}_t \sim N(\mathbf{0}, \Sigma_t)$, and maximize this Gaussian likelihood (a function of λ and the parameters in $\boldsymbol{\mu}_t$.) The function **EWMAvol** can be used for this.

BEKK. Introduced by Engle and Kroner (1995); tries to mimic a GARCH(m, s), but the curse of dimensionality prevents its usage beyond the simple $m = 1 = s$ case, so that the BEKK(1, 1) model is:

$$\Sigma_t = A_0 A_0' + A_1 (\mathbf{a}_{t-1} \mathbf{a}_{t-1}') A_1' + B_1 \Sigma_{t-1} B_1',$$

where A_0 is a lower triangular matrix, and A_1 and B_1 are unrestricted square matrices.

Summary Remarks:

- Pros: positive definite.
- Cons: Many parameters; dynamic relations require further study.
- Estimation: quasi-MLE (QMLE) similar to GARCH; can use function `BEKK11` ($k = 2$ and $k = 3$ dimensional cases only).

11.3 Dynamic Conditional Correlation (DCC) Models

These are the most popular models, and involve a three-step process. We start once again from the WN residuals in (1).

Step 1. Fit a univariate GARCH to each component residual series $\{\hat{a}_{i,t}\}$ in order to obtain the volatility estimate $\hat{\sigma}_{ii,t}$, and compute the standardized innovations:

$$\hat{\eta}_{i,t} = \hat{a}_{i,t} / \sqrt{\hat{\sigma}_{ii,t}}.$$

As vectors, we then have:

$$\hat{\boldsymbol{\eta}}_t = \hat{V}_t^{-1} \hat{\mathbf{a}}_t, \quad \text{and} \quad \hat{\mathbf{a}}_t = \hat{V}_t \hat{\boldsymbol{\eta}}_t.$$

Step 2. Use a joint dependence model on $\hat{\boldsymbol{\eta}}_t$ to capture the time evolution of the conditional correlation R_t of $\hat{\mathbf{a}}_t$ (and ultimately the evolution of $\Sigma_t = V_t R_t V_t$). Let $\boldsymbol{\theta}$ denote the parameters of this model.

Step 3. With $\ell_t(\boldsymbol{\theta})$ the log of the pdf of $\hat{\boldsymbol{\eta}}_t$, estimate $\boldsymbol{\theta}$ by maximizing the (conditional) log-likelihood:

$$\ell(\boldsymbol{\theta}) = \sum_{t=1}^T \ell_t(\boldsymbol{\theta}).$$

A sensible choice for the pdf of $\boldsymbol{\eta}_t$ is a multivariate t_ν . The pdf of $\mathbf{x} \sim t_\nu(\boldsymbol{\mu}, \Sigma)$, where $\boldsymbol{\mu}$ and Σ are location and scale parameters, is proportional to:

$$f_\nu(\mathbf{x}; \boldsymbol{\mu}, \Sigma) \propto |\Sigma|^{-1/2} \left[1 + \frac{(\mathbf{x} - \boldsymbol{\mu})' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})}{\nu} \right]^{-(k+\nu)/2}.$$

If the degrees of freedom $\nu > 2$, then:

$$\mathbb{E}(\mathbf{x}) = \boldsymbol{\mu}, \quad \text{and} \quad \mathbb{V}(\mathbf{x}) = \frac{\nu}{\nu - 2} \Sigma.$$

However, since $\boldsymbol{\eta}_t$ is standardized,

$$\mathbb{E}(\boldsymbol{\eta}_t) = \mathbf{0}, \quad \text{and} \quad \mathbb{V}(\boldsymbol{\eta}_t) = R_t,$$

we should use the t_ν^* , a standard t_ν whose pdf is proportional to:

$$f_\nu^*(\boldsymbol{\eta}_t; \boldsymbol{\mu}_t = \mathbf{0}, \Sigma_t = R_t) \propto |R_t|^{-1/2} \left[1 + \frac{\boldsymbol{\eta}_t' R_t^{-1} \boldsymbol{\eta}_t}{\nu - 2} \right]^{-(k+\nu)/2}.$$

For modeling R_t , three types of DCC are available in the literature.

- (i) **Engle (2002)**. Let $W_t = \text{Diag}(\sqrt{Q_{11,t}}, \dots, \sqrt{Q_{kk,t}})$ be a diagonal matrix whose elements are taken from those of Q_t , defined recursively as:

$$\begin{aligned} Q_t &= (1 - \theta_1 - \theta_2)R_0 + \theta_1 Q_{t-1} + \theta_2 \boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}' \\ R_t &= W_t^{-1} Q_t W_t^{-1}, \end{aligned}$$

where $0 \leq \theta_i$ and $\theta_1 + \theta_2 < 1$, and R_0 is the sample correlation matrix of the $\{\hat{\boldsymbol{\eta}}_t\}$.

- (ii) **Tse and Tsui (2002)**. Define R_t recursively as:

$$R_t = (1 - \theta_1 - \theta_2)R_0 + \theta_1 R_{t-1} + \theta_2 \Psi_{t-1},$$

where the θ_i and R_0 are as above, and Ψ_{t-1} is the (local) sample correlation matrix of $\{\mathbf{a}_{t-1}, \dots, \mathbf{a}_{t-m}\}$ for a pre-specified positive integer $m > k$ (acts as a smoothing parameter, larger m give smoother correlations).

- (iii) **Van der Weide (2003)**. Decompose $R_t = X_t' X_t$, where X_t is an upper triangular matrix parametrized in terms of sines and cosines of a vector of angles $\boldsymbol{\theta}_t$ evolving via a DCC-type equation:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_0 + \lambda_1 \boldsymbol{\theta}_{t-1} + \lambda_2 \boldsymbol{\theta}_{t-1}^*,$$

where $\boldsymbol{\theta}_{t-1}^*$ is a local estimate of the angles using data $\{\hat{\boldsymbol{\eta}}_{t-1}, \dots, \hat{\boldsymbol{\eta}}_{t-m}\}$ for some $m > 1$, the λ_i are non-negative numbers satisfying $0 < \lambda_1 + \lambda_2 < 1$, and $\boldsymbol{\theta}_0$ denotes the initial values of the angles.

Summary Remarks:

- For the univariate GARCH fits in Step 1: use function **dccPre**.
- DCC models are extremely simple with only two parameters to estimate. The function **dccFit** allows the fitting of models in (i)–(ii), while the model in (iii) can be fitted via **mtCopula** (in the context of copula models to be discussed next).
- Goodness of fit checks tend to reject DCC models. The function **MCHdiag** carries out 4 portmanteau tests to diagnose the fit.

11.4 Copula-Based Models

Copulas: Tools for modeling the (joint) dependence structure among $k \geq 2$ random variables; from the Latin *copulare* (to connect or join). Thus the copula connects the marginals into a joint distribution.

Def: A k -dimensional copula is a function:

$$C(u_1, \dots, u_k) : [0, 1]^k \mapsto [0, 1],$$

which is a k -dimensional cdf with $\text{Unif}[0, 1]$ marginals: $U_1, \dots, U_k$.

Key Idea ($k = 2$): For any r.v.s $X_i = F_i^{-1}(U_i)$, where $U_i \sim \text{Unif}[0, 1]$ and F_i^{-1} is the (generalized) inverse cdf of X_i , then

$$\begin{aligned} P(X_1 \leq x_1, X_2 \leq x_2) &= P(F_1^{-1}(U_1) \leq x_1, F_2^{-1}(U_2) \leq x_2) \\ &= P(U_1 \leq F_1(x_1), U_2 \leq F_2(x_2)) := C(u_1, u_2), \end{aligned}$$

where $u_1 = F_1(x_1)$ and $u_2 = F_2(x_2)$. Thus $C(u_1, u_2)$ describes the dependence structure between X_1 and X_2 , stripped of the marginals, w.r.t. standard uniforms (which comprise the reference distribution). Since $C(u_1, u_2)$ is invariant to the marginals, it can be used to connect any two marginals.

Sklar's Theorem ($k = 2$ case): Let $F(x_1, x_2)$ be a 2-dim cdf with marginals $F_1(x_1)$ and $F_2(x_2)$. Then there exists a copula $C(u_1, u_2)$ such that $F(x_1, x_2) = C(F_1(x_1), F_2(x_2))$, and it's unique if $F(x_1, x_2)$ is continuous. Conversely, if $C(\cdot, \cdot)$ is a copula and F_1 and F_2 are univariate cdf's, then the function $C(F_1(x_1), F_2(x_2))$ is a joint cdf with marginals F_1 and F_2 .

Note: The main consequence of Sklar's Theorem is that given $F(x_1, x_2)$, the underlying (implied) copula cdf (which always exists) is:

$$C(u_1, u_2) = F(F_1^{-1}(u_1), F_2^{-1}(u_2)).$$

The corresponding copula pdf, which only exists if $C(u_1, u_2)$ is differentiable, is:

$$c(u_1, u_2) = \frac{\partial^2 C(u_1, u_2)}{\partial u_1 \partial u_2}.$$

Bounds on Copulas: Recalling that for $U \sim \text{Unif}[0, 1]$, $F(u) = uI(0 \leq u \leq 1)$, we have the following bounds, which can be visualized as surfaces in the unit hypercube, and equated with degrees of independence:

- Independence (U_1 and U_2 are unrelated): If U_1 and U_2 are independent, then

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2) = P(U_1 \leq u_1)P(U_2 \leq u_2) = u_1 u_2.$$

- Comonotonic (U_1 and U_2 are the same): If U_1 and U_2 are the same, $U_1 = U_2 \equiv U$, so that:

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2) = P(U \leq u_1, U \leq u_2) = P(U \leq \min(u_1, u_2))$$

- Countermonotonic (U_1 and U_2 are opposite): If U_1 and U_2 are the opposite, $U_2 = 1 - U_1$, so that:

$$\begin{aligned} C(u_1, u_2) &= P(U_1 \leq u_1, U_2 \leq u_2) = P(U \leq u_1, 1 - U \leq u_2) = P(U \leq u_1, U \geq 1 - u_2) \\ &= \begin{cases} u_1 + u_2 - 1, & u_1 + u_2 > 1, \\ 0, & u_1 + u_2 \leq 1. \end{cases} \end{aligned}$$

Here we explore a copula-based model for the joint dependence of $\hat{\eta}_{i,t}$ in order to capture the time evolution of R_t in Step 2 of §11.3. Denote by $f_{\nu,R}^*(\mathbf{x})$ the standard t_ν pdf as above, which is proportional to:

$$f_{\nu,R}^*(\mathbf{x}) \propto |R|^{-1/2} \left[1 + \frac{\mathbf{x}' R^{-1} \mathbf{x}}{\nu - 2} \right]^{-(k+\nu)/2},$$

and let $f_\nu^*(x)/F_\nu^*(x)$ denote the pdf/cdf of a univariate standard t_ν . If the data are $\mathbf{x} = (x_1, \dots, x_k)'$, then the pdf of the (std.) t -copula evaluated at $\mathbf{u} \in [0, 1]^k$ is:

$$c_{\nu,R}^t(\mathbf{u}) = \frac{f_{\nu,R}^*(\mathbf{x})}{\prod_{i=1}^k f_\nu^*(x_i)}, \quad \text{where } x_i = F_\nu^{*-1}(u_i).$$

Now maximize this copula likelihood for the standardized innovations, i.e., set $x_i = \hat{\eta}_{i,t}$, and use a model for the time evolution of $R \equiv R_t$ (Steps 2–3 of §11.3). One of the DCC models could be used here, but the function

mtCopula fits a t -copula by parametrizing R_t via angles, and the angles evolve via the model of Van der Weide (2003).

Note: This t -copula likelihood is not multivariate t ; it's called a meta- t_ν .

Ex: The left panels of Figure 1 display lag 1 scatterplots (Z_{t-1}, Z_t) , for 10,000 simulated values from $Z_t \sim \text{ARCH}(1)$ with $\alpha_0 = 1$ and $\alpha_1 = 0.5$ (top panel), and from $Z_t \sim \text{GARCH}(1,1)$ with $\alpha_0 = 1$, $\alpha_1 = 0.1$, and $\beta_1 = 0.8$ (bottom panel). The right panels are nonparametric estimates of the corresponding implied copula densities.

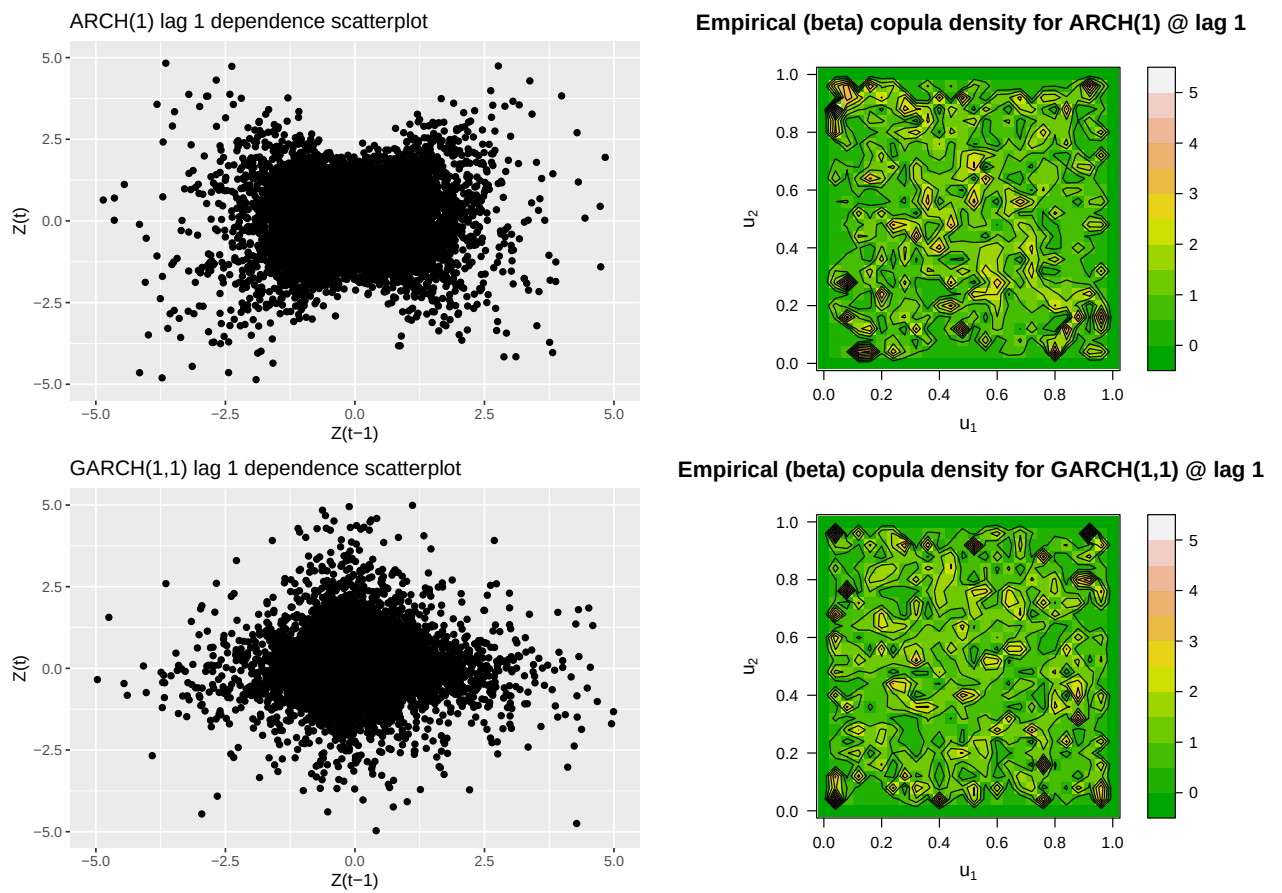


Figure 1: ARCH and GARCH implied copula illustration.

11.5 Examples

Example 1: (Example 7.7.2 in MTS, Tsay, 2014) Dataset `m-ibmspko-6111.txt` contains the monthly values of IBM stock, S&P composite index, and Coca Cola stock, from January 1961 to December 2011. After converting to log returns, we fit the multivariate t_ν DCC model of §11.3. (Fig. 2 shows the resulting pairwise R_t .)

```
> da=read.table("m-ibmspko-6111.txt",header=T)
> rtn=log(da[,2:4]+1)
> m1=dccPre(rtn,include.mean=T,p=0)
Sample mean of the returns: 0.00772774 0.005023909 0.01059521
Component: 1
Estimates: 0.000419 0.126739 0.788307
se.coef : 0.000162 0.035405 0.055645
t-value : 2.593448 3.57973 14.16662
Component: 2
Estimates: 9e-05 0.127725 0.836053
se.coef : 4.1e-05 0.03084 0.031723
t-value : 2.20126 4.141592 26.35486
Component: 3
Estimates: 0.000256 0.098705 0.830358
se.coef : 8.5e-05 0.022361 0.033441
t-value : 3.015321 4.414112 24.83088

> names(m1)
[1] "marVol" "sresi" "est" "se.coef"

> rtn1=m1$sresi
> Vol=m1$marVol
> m2=dccFit(rtn1)
Estimates: 0.8088086 0.04027318 7.959013
st.errors: 0.1491655 0.02259863 1.135882
```

```
t-values:  5.422222  1.782107  7.006898
```

```
> names(m2)
```

```
[1] "estimates" "Hessian" "rho.t"
```

```
> S2.t = m2$rho.t
```

```
> m3=dccFit(rtn1,type="Engle")
```

```
Estimates: 0.9126634 0.04530917 8.623668
```

```
st.errors: 0.0294762 0.01273911 1.332381
```

```
t-values: 30.96272  3.556697  6.472376
```

```
> S3.t=m3$rho.t
```

```
> MCHdiag(rtn1,S2.t)
```

```
Test results:
```

```
Q(m) of et:
```

```
Test and p-value: 20.74262 0.02296152
```

```
Rank-based test:
```

```
Test and p-value: 30.20662 0.0007924436
```

```
Qk(m) of epsilon_t:
```

```
Test and p-value: 132.423 0.002425885
```

```
Robust Qk(m):
```

```
Test and p-value: 109.9671 0.0750157
```

```
> MCHdiag(rtn1,S3.t)
```

```
Test results:
```

```
Q(m) of et:
```

```
Test and p-value: 20.02958 0.02897411
```

```
Rank-based test:
```

```
Test and p-value: 27.61638 0.002078829
```

```
Qk(m) of epsilon_t:
```

```
Test and p-value: 131.982 0.002625755
```

```
Robust Qk(m):
```

Test and p-value: 111.353 0.06307334

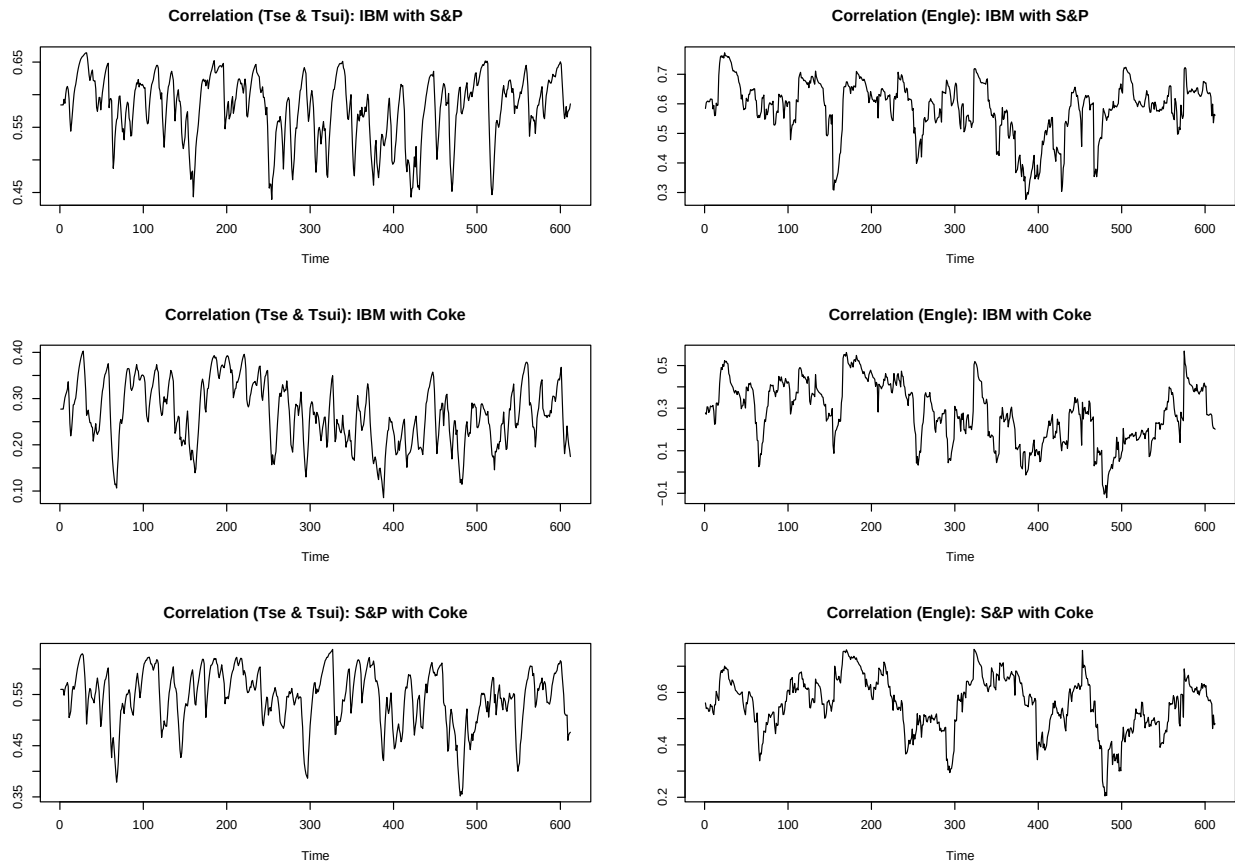


Figure 2: DCC model with t_ν for the 3 series: IBM, S&P, Coke.

Example 2: (Example 7.5 in MTS, Tsay, 2014) Analyzes again `m-ibmspko-6111.txt`: the monthly values of IBM stock, S&P composite index, and Coca Cola stock, from January 1961 to December 2011. Here we fit the meta- t_ν copula DCC model of §11.4 to the log returns, with the Van der Weide (2003) dynamic correlation structure. (Fig. 3 shows the individual volatility estimates, and pairwise R_t .)

```
> da=read.table("m-ibmspko-6111.txt",header=T)
> rtn=log(da[,-1]+1)
> m1=dccPre(rtn,cond.dist="std")
Sample mean of the returns: 0.00772774 0.005023909 0.01059521
Component: 1
Estimates: 0.000388 0.115626 0.805129 9.209269
se.coef : 0.000177 0.036827 0.059471 3.054817
t-value : 2.195398 3.139719 13.5382 3.014671
Component: 2
Estimates: 0.00012 0.130898 0.814531 7.274928
se.coef : 5.7e-05 0.037012 0.046044 1.913331
t-value : 2.102768 3.536655 17.69028 3.802232
Component: 3
Estimates: 0.000216 0.104706 0.837217 7.077138
se.coef : 8.9e-05 0.028107 0.037157 1.847528
t-value : 2.437323 3.725341 22.53208 3.830599

> names(m1)
[1] "marVol" "sresi" "est" "se.coef"

> Vol=m1$marVol; eta=m1$sresi
> m2=mtCopula(eta,0.8,0.04)
Lower limits: 5.1 0.2 1e-04 0.7564334 1.031269 0.8276595
Upper limits: 20 0.95 0.04999999 1.040096 1.417994 1.138032
estimates: 15.38215 0.88189 0.034025 0.919724 1.225322 1.058445
std.errors: 8.222771 0.05117 0.011733 0.041357 0.055476 0.051849
t-values: 1.870677 17.2341 2.899996 22.23883 22.08729 20.41412
Alternative numerical estimates of se:
st.errors: 5.477764 0.051033 0.011714 0.041370 0.055293 0.050793
t-values: 2.808107 17.28091 2.904679 22.23173 22.16072 20.83839

> ### Estimate all parameters (including theta_0)
> names(m2)
[1] "estimates" "Hessian" "rho.t" "theta.t"
```

```

> MCHdiag(eta,m2$rho.t)
Test results:
Q(m) of et:
Test and p-value: 19.30177 0.03659304
Rank-based test:
Test and p-value: 27.03262 0.002573576
Qk(m) of epsilon_t:
Test and p-value: 125.9746 0.007387423
Robust Qk(m):
Test and p-value: 107.4675 0.1011374

> ### Restimate, but fix theta_0 (based on the sample correlations)
> m3=mtCopula(eta,0.8,0.04,include.th0=F)
Value of angles:
[1] 0.9455418 1.2890858 1.0345744
Lower limits: 5.1 0.2 1e-05
Upper limits: 20 0.95 0.0499999
estimates: 14.87427 0.8778 0.03365157
std.errors: 7.959968 0.053013 0.011951
t-values: 1.868635 16.55824 2.815811
Alternative numerical estimates of se:
st.errors: 5.49568 0.0529896 0.01191378
t-values: 2.70654 16.56551 2.824592

```

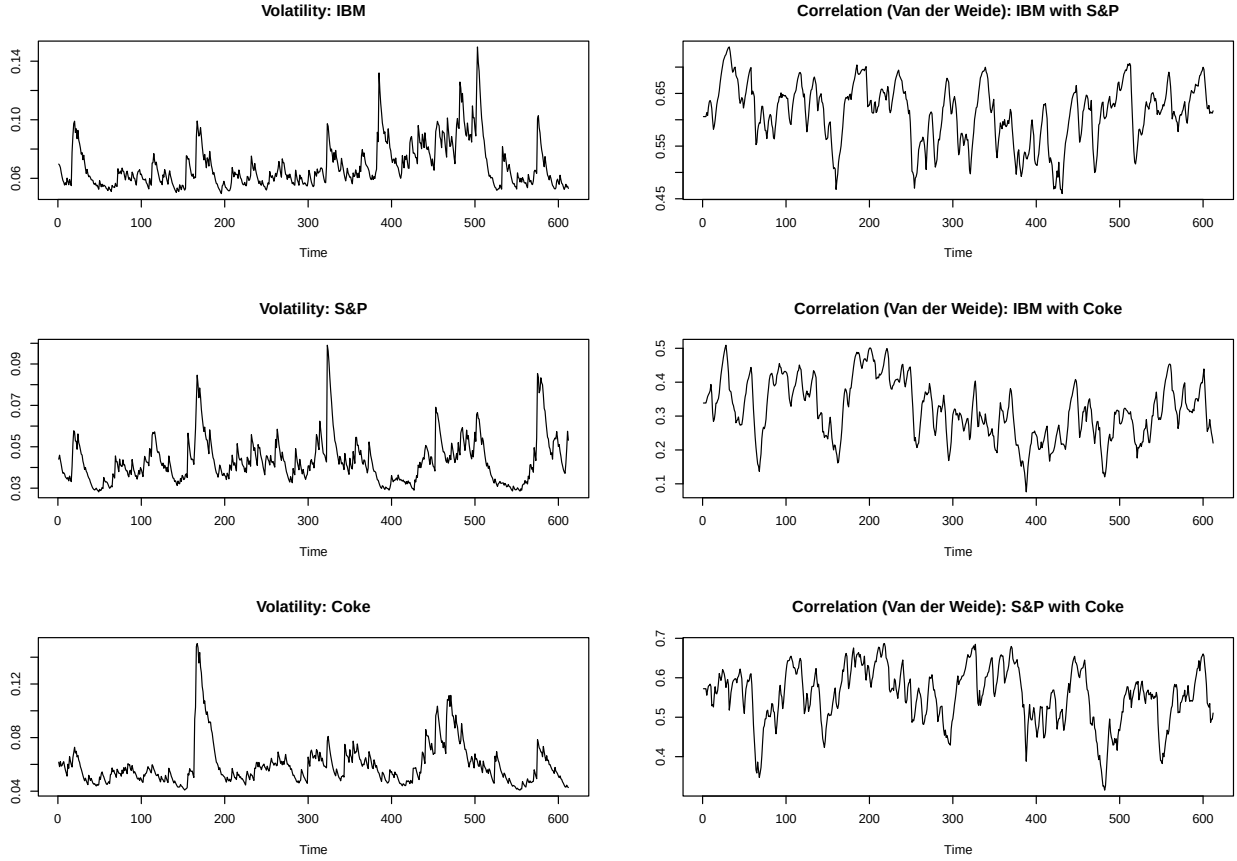



Figure 3: Volatilities and DCC model with meta- t_ν copula for the 3 series: IBM, S&P, Coke.

11.6 VaR and ES via Multivariate Modeling

From our k -dim volatility models:

$$\mathbf{r}_t = \boldsymbol{\mu}_t + \mathbf{a}_t, \quad \mathbf{a}_t = \Sigma_t^{1/2} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim (\mathbf{0}, I_k),$$

so that conditionally on F_{t-1} : $\mathbf{r}_t \sim (\boldsymbol{\mu}_t, \Sigma_t)$. Thus, for a portfolio $z_t = \mathbf{w}'\mathbf{r}_t$, where the weights vector $\mathbf{w}' = (w_1, \dots, w_k)$ is composed of deterministic elements, the conditional mean and variance of z_t are:

$$z_t | F_{t-1} \sim (\mathbf{w}'\boldsymbol{\mu}_t, \mathbf{w}'\Sigma_t\mathbf{w}).$$

Most common cases are when $\mathbf{r}_t$ is (conditionally) normal or t (both members of the Elliptically Contoured family of distributions).

- Conditionally Normal: If $\mathbf{r}_t \sim N(\boldsymbol{\mu}_t, \Sigma_t)$, then

$$z_t \sim N(\mathbf{w}'\boldsymbol{\mu}_t, \mathbf{w}'\Sigma_t\mathbf{w}).$$

- Conditionally t_ν^* : If $\mathbf{r}_t \sim t_\nu^*(\boldsymbol{\mu}_t, \Sigma_t)$, a standard t_ν with (conditional) mean and variance $\boldsymbol{\mu}_t$ and Σ_t ($\nu \geq 3$), then

$$z_t \sim t_\nu^*(\mathbf{w}'\boldsymbol{\mu}_t, \mathbf{w}'\Sigma_t\mathbf{w}).$$

Proofs: These equations follow from the fact that (conditionally):

$$\boldsymbol{\eta}_t = V_t^{-1}\mathbf{a}_t \sim (\mathbf{0}, R_t) \implies \mathbf{a}_t = V_t\boldsymbol{\eta}_t \sim (\mathbf{0}, V_t R_t V_t' = \Sigma_t) \implies \mathbf{r}_t = \boldsymbol{\mu}_t + \mathbf{a}_t \sim (\boldsymbol{\mu}_t, \Sigma_t).$$

These results can now be applied using the univariate formulas for VaR and ES from Lecture 9, given forecasts for $\boldsymbol{\mu}_t$ and Σ_t (e.g., 1-step ahead). While forecasts of $\boldsymbol{\mu}_t$ are straightforward (e.g., from a fitted VAR), the forecasts for Σ_t are more tricky....

For the Tse and Tsui (2002) model, the forecasts of R_t are similar to AR/ARCH:

$$R_t = (1 - \theta_1 - \theta_2)R_0 + \theta_1 R_{t-1} + \theta_2 \Psi_{t-1} \implies R_{t+1} = (1 - \theta_1 - \theta_2)R_0 + \theta_1 R_t + \theta_2 \Psi_t.$$

Since $\{R_0, R_t, \Psi_t\} \in F_t$, the best 1-step predictor follows the model:

$$\hat{R}_t(1) = \mathbb{E}(R_{t+1}|F_t) = (1 - \theta_1 - \theta_2)R_0 + \theta_1 R_t + \theta_2 \Psi_t,$$

and similarly for multi-step predictors.